

08/08/24

Chapter-9 (B %)

Turing Machine (17 M) (4 Turing M. each 4M)

- Most powerful model.
- Invented by Alan Turing in 1936.
- It accepts recursively enumerable language.
- Solves the problem or functions which is not solved by PDA and FA.
- Some functions that are not stopped by Turing Machine, goes to Undecidable Model.

Undecidable



Turing Machine (Recursive enumerable)

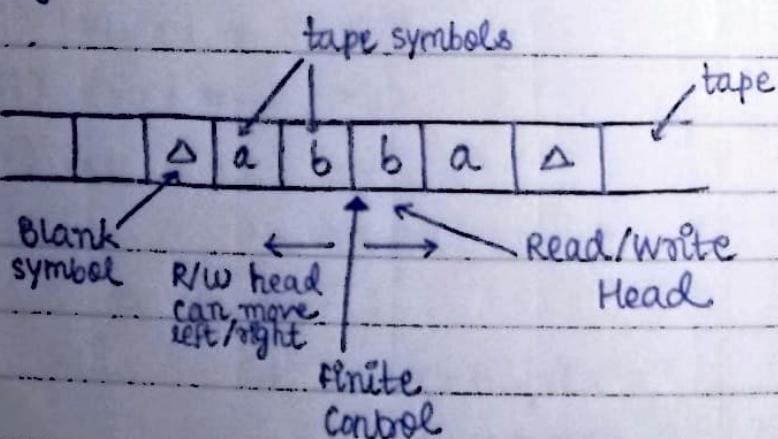


PDA (Context Free language)



FA (Regular language)

→ Turing Machine Structure :-



→ It consists of 7 tuples -
 $(Q, \Sigma, \Gamma, \delta, q_0, B, F)$

Q = set of states

Σ = Input symbols

Γ = Finite state of tape symbols

δ = transition function

B = Blank symbol

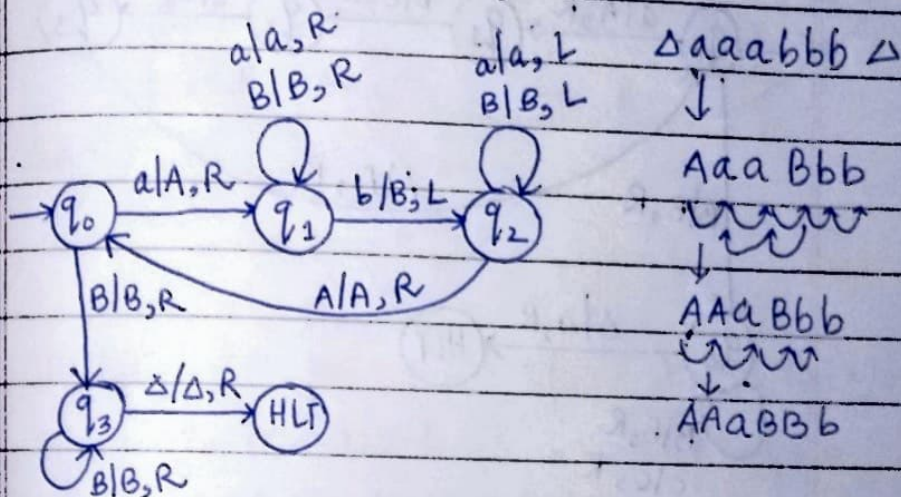
q_0 = Initial state

F = Set of final states

(1) Design TM for $L = \{a^n b^n \mid n \geq 1\}$

→

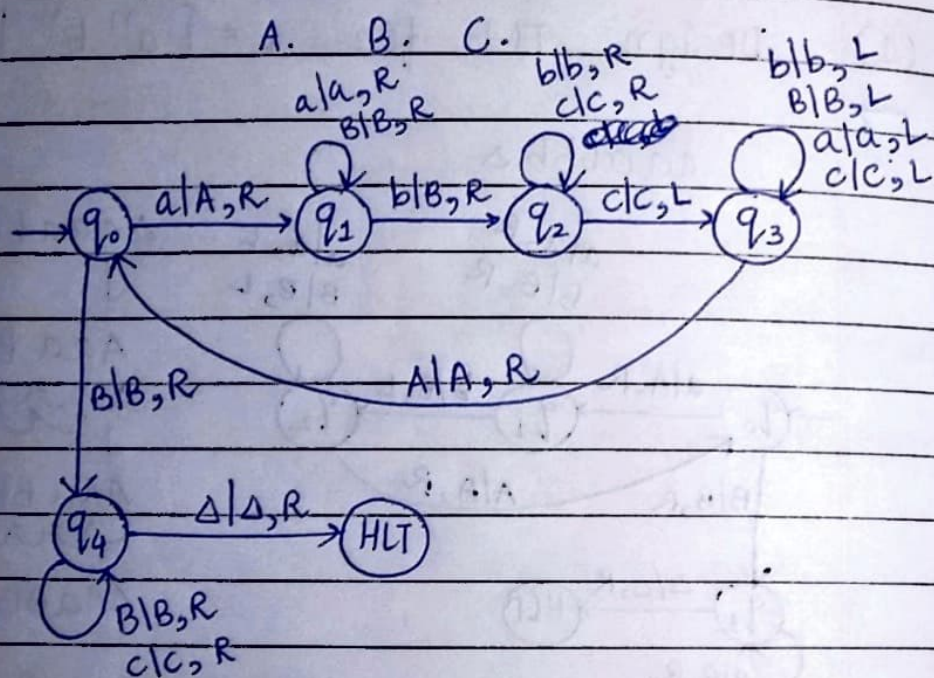
aaabbbb Δ



States	a	b	A	B	Δ
q_0	q_1, A, R	-	-	q_3, B, R	-
q_1	q_1, a, R	q_2, B, L	-	q_1, B, R	-
q_2	q_2, a, L	-	q_0, A, R	q_2, B, L	-
q_3	-	-	-	q_3, B, R	HLT, Δ, R

Trace the string - 'aabb Δ '
 $(q_0, aabb\Delta) \vdash (q_1, Aabb\Delta)$
 $\vdash (q_2, ABb\Delta)$
 $\vdash (q_0, ABb\Delta)$
 $\vdash (q_1, AABb\Delta)$
 $\vdash (q_2, AABb\Delta)$
 $\vdash (q_3, AABb\Delta)$

(2) Design TM for $L = \{a^n b^n c^n \mid n \geq 1\}$
 $\rightarrow$ aaabbbccc



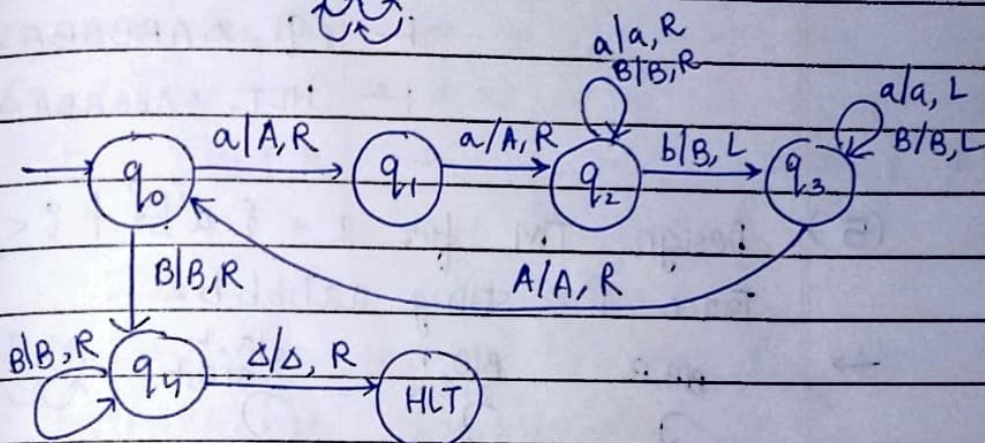
States	a	b	A	B	c	C	Δ
q_0	q_1, A, R	-	-	q_4, B, R	-	-	-
q_1	q_1, a, R	q_2, B, R	-	q_1, b, R	-	-	-
q_2	-	q_2, b, R	-	-	-	-	-
q_3	q_3, a, L	q_3, b, L	q_0, A, R	q_3, B, L	q_3, C, L	-	-
q_4	-	-	-	q_4, B, R	-	-	HLT Δ, R

(3)

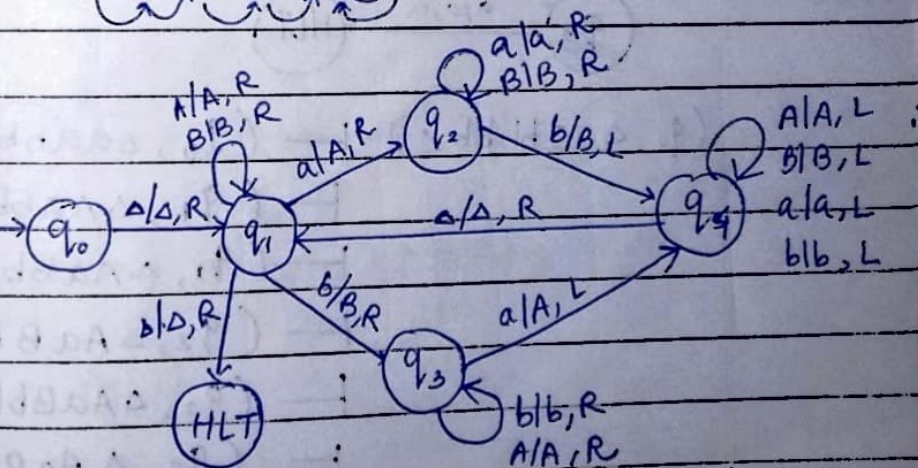
Design TM for $L = \{a^n b^n \mid n \geq 1\}$ $\Rightarrow$ for $n=2$,

(4 marks)

aaaa bb Δ

AAAABb Δ


(4)

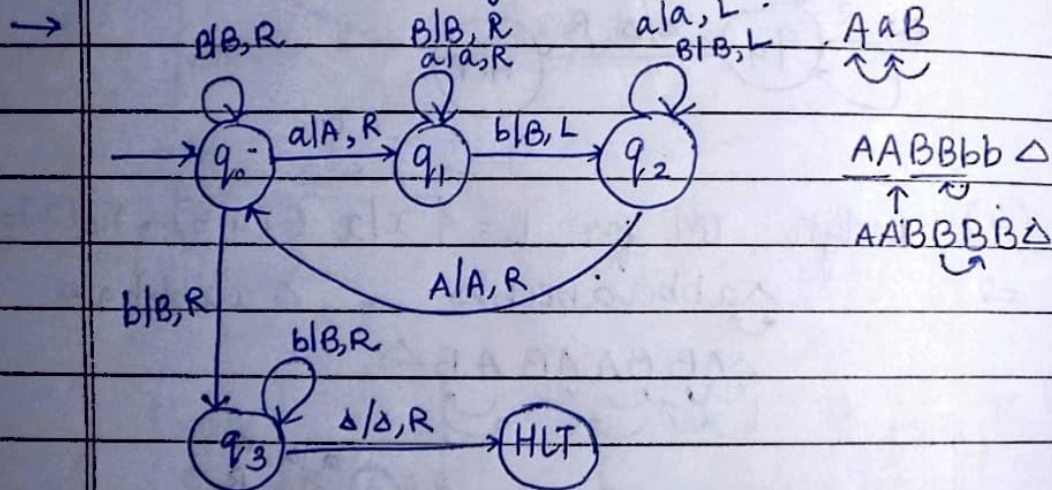
Design TM for $L = \{x \mid x \in (a,b), n_a(x) = n_b(x)\}$ $\Rightarrow$
 $\Delta abbaabab \Delta$, $\Delta aabbba \Delta$
 $\Delta ABBAABAB \Delta$


Trace the string 'sabbabab' - :

$(q_0, sabbabab\Delta) \vdash (q_1, sabbabab\Delta)$
 $\vdash (q_2, sAbbabab\Delta)$
 $\vdash (q_4, sABbabab\Delta)$
 $\vdash (q_1, sABbabab\Delta)$
 $\vdash (q_3, sABbabab\Delta)$
 $\vdash (q_4, sABBAbab\Delta)$
 $\vdash (q_1, sABBAbab\Delta)$
 $\vdash (q_3, sABBAba\Delta)$
 $\vdash (q_4, sABBAba\Delta)$
 $\vdash (q_1, sABBAba\Delta)$
 $\vdash (HLT, sABBAba\Delta)$

(5) Design TM for $L = \{a^i b^j \mid i < j, i \geq 0\}$

Trace the string aabbbb Δ .



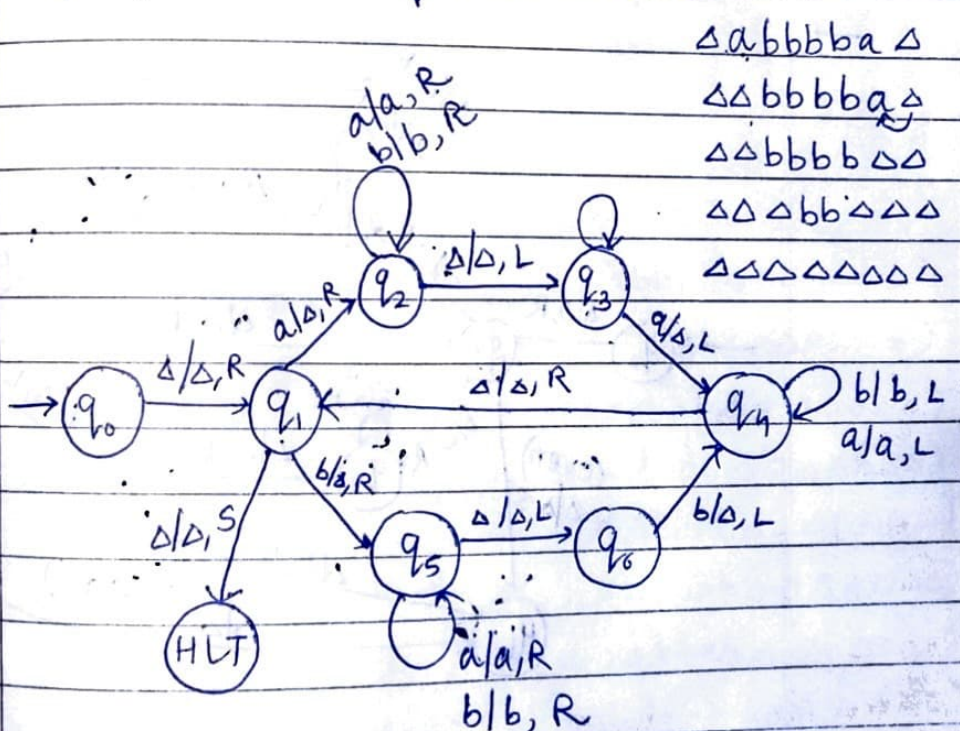
$(q_0, saabbbb\Delta) \vdash (q_0, saabbbb\Delta)$
 $\vdash (q_1, sAabbbb\Delta)$
 $\vdash (q_1, sAabbbb\Delta)$
 $\vdash (q_2, sAABbbb\Delta)$
 $\vdash (q_2, sAABbbb\Delta)$
 $\vdash (q_0, sAABbbb\Delta)$

$\vdash (q_1, \Delta AABbbb\Delta)$
 $\vdash (q_1, \Delta AABbbb\Delta)$
 $\vdash (q_2, \Delta AABBBb\Delta)$
 $\vdash (q_0, \Delta AABBBb\Delta)$
 $\vdash (q_3, \Delta AABBBb\Delta)$
 $\vdash (q_3, \Delta AABBBB\Delta)$
 $\vdash (q_{HLT}, \Delta AABBBB\Delta)$

States	a	b	A	B	HLT
q_0	(q_1, A, R)	(q_3, B, R)	—	(q_0, B, R)	—
q_1	(q_1, a, R)	(q_2, B, L)	—	(q_1, B, R)	—
q_2	(q_2, a, L)	—	(q_0, A, R)	(q_2, B, L)	—
q_3	—	(q_3, B, R)	—	—	(q_{HLT}, Δ, R)
q_{HLT}	—	—	—	—	—

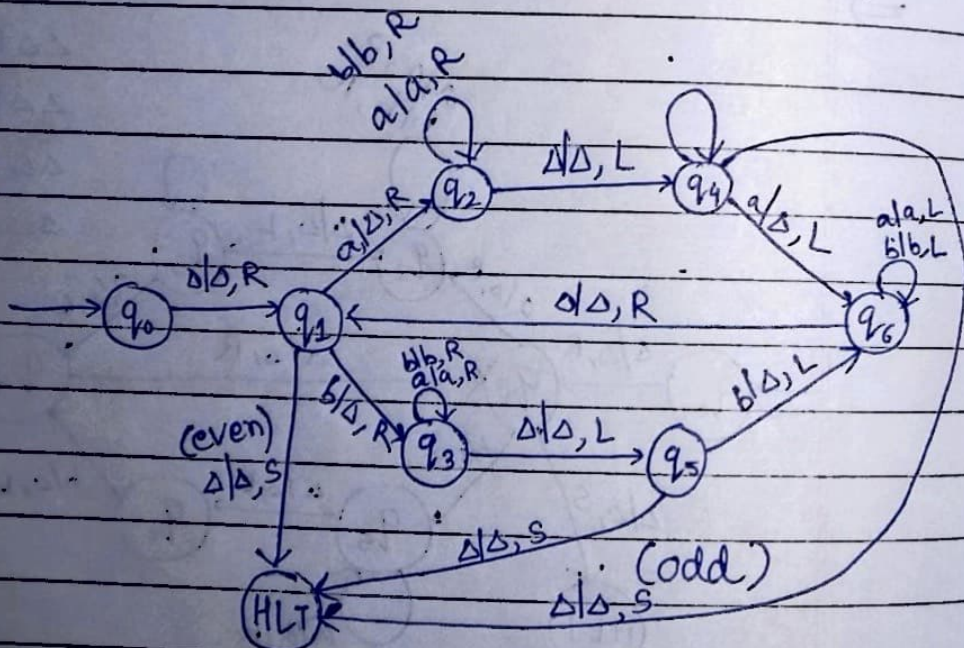
(6) Design TM for $t = \frac{1}{2}$, even length palindrome from (a, b)

$\Rightarrow$



(8)

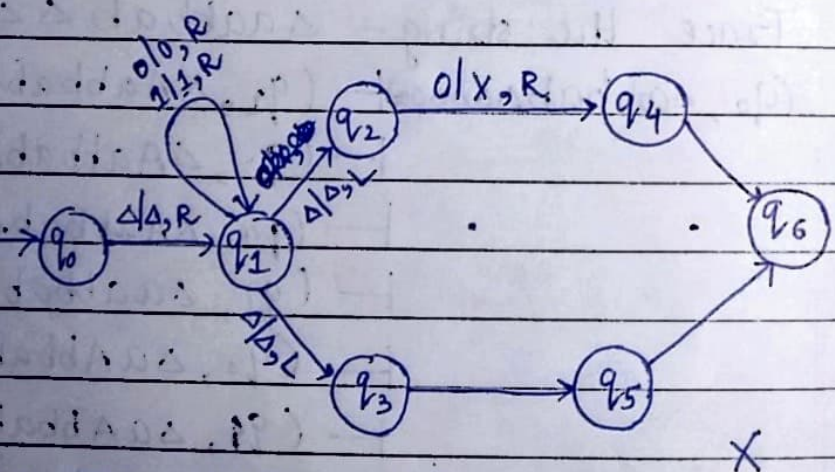
→



- ⊢ (q₆, Δ a a B b a b Δ a a Δ Δ Δ Δ)
- ⊢ (q₄, Δ a a B b a b Δ a a b Δ Δ Δ)
- ⊢ (q₁, Δ a a b b a b Δ a a b Δ Δ Δ)
- ⊢ (q₅, Δ a a b B a b Δ a e b Δ Δ Δ)
- ⊢ (q₆, Δ a a b B a b Δ a a b Δ Δ Δ)
- ⊢ (q₄, Δ a a b B a b Δ a a b b Δ Δ Δ)
- ⊢ (q₁, Δ a a b b a b Δ a a b b Δ Δ Δ)
- ⊢ (q₂, Δ a a b b A b Δ a a b b Δ Δ Δ)
- ⊢ (q₃, Δ a a b b A b Δ a a b b Δ Δ Δ)
- ⊢ (q₄, Δ a a b b A b Δ a a b b a Δ Δ)
- ⊢ (q₁, Δ a a b b a b Δ a a b b a Δ Δ)
- ⊢ (q₅, Δ a a b b a B Δ a a b b a Δ Δ)
- ⊢ (q₆, Δ a a b b a B Δ a a b b a Δ Δ)
- ⊢ (q₄, Δ a a b b a B Δ a a b b a b Δ Δ)
- ⊢ (q₁, Δ a a b b a b Δ a a b b a b Δ Δ)
- ⊢ (HLT, Accepted).

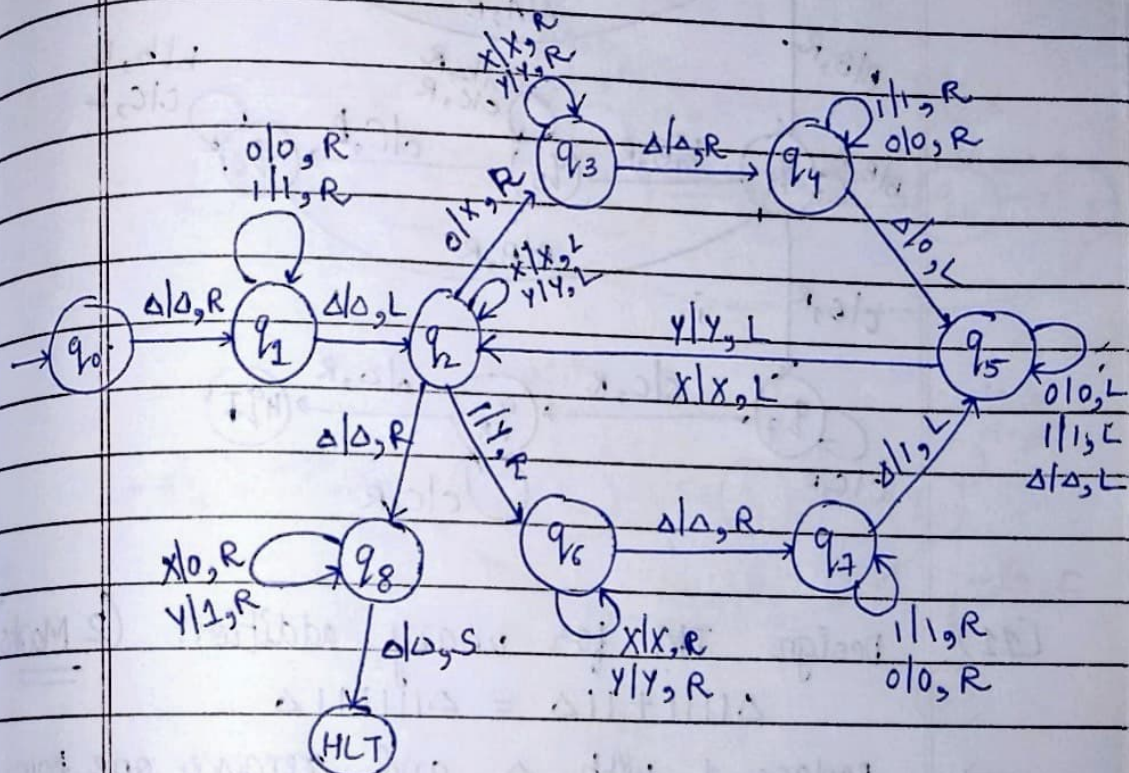
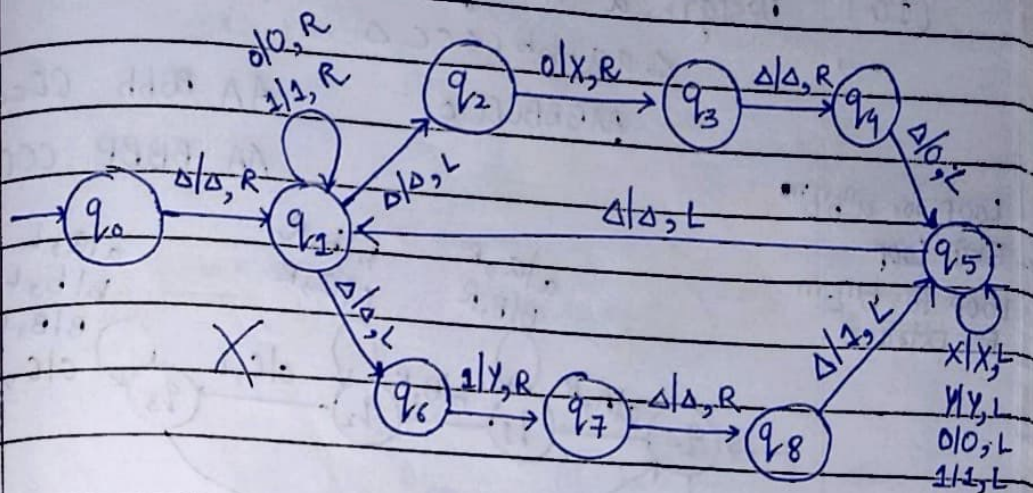
(9) Design TM for reverse a string from $\Sigma = \{0,1\}$.

Δ 0 1 1 0 1 Δ Δ Δ Δ Δ Δ 0 → x, 1 → y
Δ 0 1 1 0 y Δ 1 Δ Δ Δ Δ



- States
- q₀
 - q₁
 - q₂
 - q₃
 - q₄
 - q₅
 - q₆
 - q₇
 - q₈

Δ 0 1 1 0 y Δ 0 1 x y Δ 1 0 Δ Δ Δ Δ
 Δ 0 1 y x y Δ 1 0 1 Δ Δ Δ
 Δ 0 y y x y Δ 1 0 1 1 Δ
 Δ x y y x y Δ 1 0 1 1 Δ → Δ 0 1 1 0 1 Δ 1 0 1 1 0

 $\Sigma = \{0, 1\}$ $1 \rightarrow Y$

States	0	1	X	Y	HLT
q_0	—	—	—	—	—
q_1	$(q_1, 0, R)$	$(q_1, 1, R)$	—	—	—
q_2	(q_2, X, R)	(q_6, Y, R)	(q_2, X, L)	(q_2, Y, L)	—
q_3	—	—	(q_3, X, R)	(q_3, Y, R)	—
q_4	$(q_4, 0, R)$	$(q_4, 1, R)$	—	—	—
q_5	$(q_5, 0, L)$	$(q_5, 1, L)$	(q_2, X, L)	(q_2, Y, L)	—
q_6	—	—	(q_6, X, R)	(q_6, Y, R)	—
q_7	$(q_7, 0, R)$	$(q_7, 1, R)$	—	—	—
q_8	$(q_8, —)$	—	$(q_8, 0, R)$	$(q_8, 1, R)$	—

(10) Design a TM for $L = \{ a^i b^j c^k \mid i < j < k; i \geq 1 \}$
 $\Delta aabbcccc \Delta$

AA BB BB CCC C

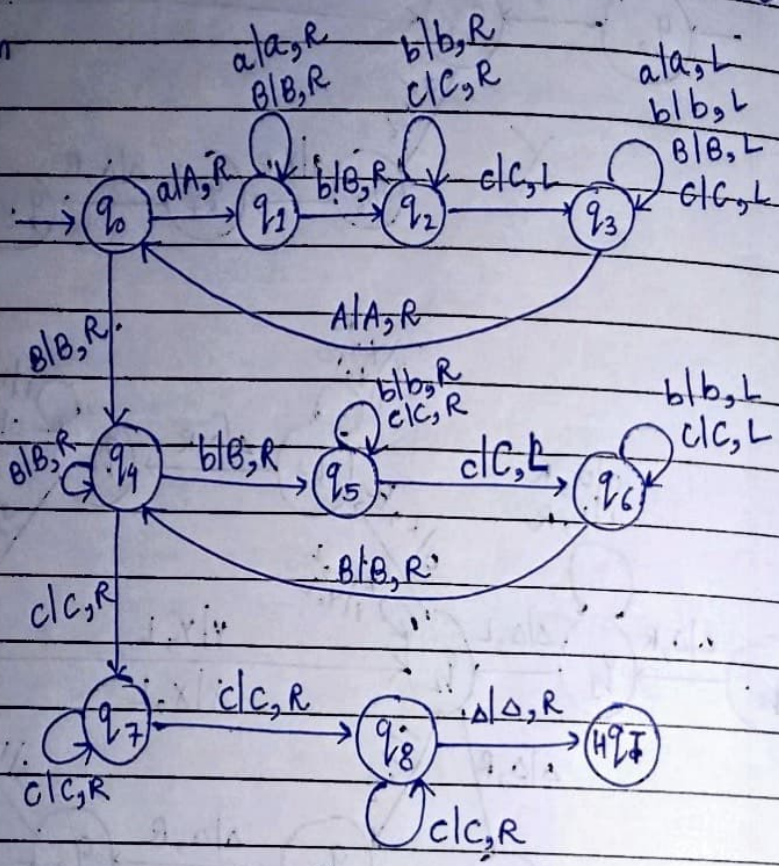
AA BBbb CCccc

AA BBBB CCCCC

Loop for $a^i b^j c^k$

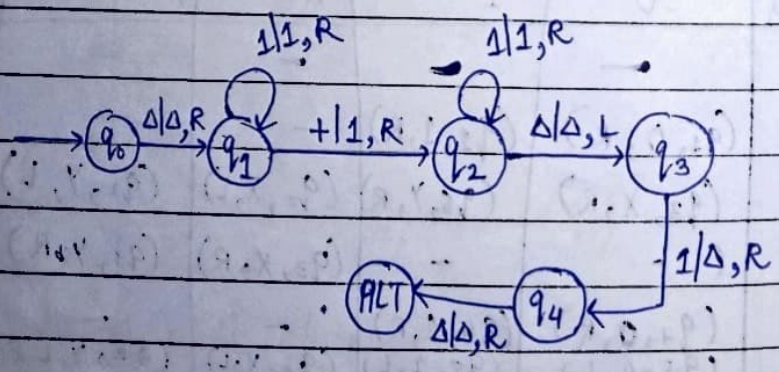
Exit loop

Loop for $b^j c^k$
For Extra c



(11) Design TM for unary addition (2 Marks)
 $\Delta 1111 + 11 \Delta = \Delta 111111 \Delta$

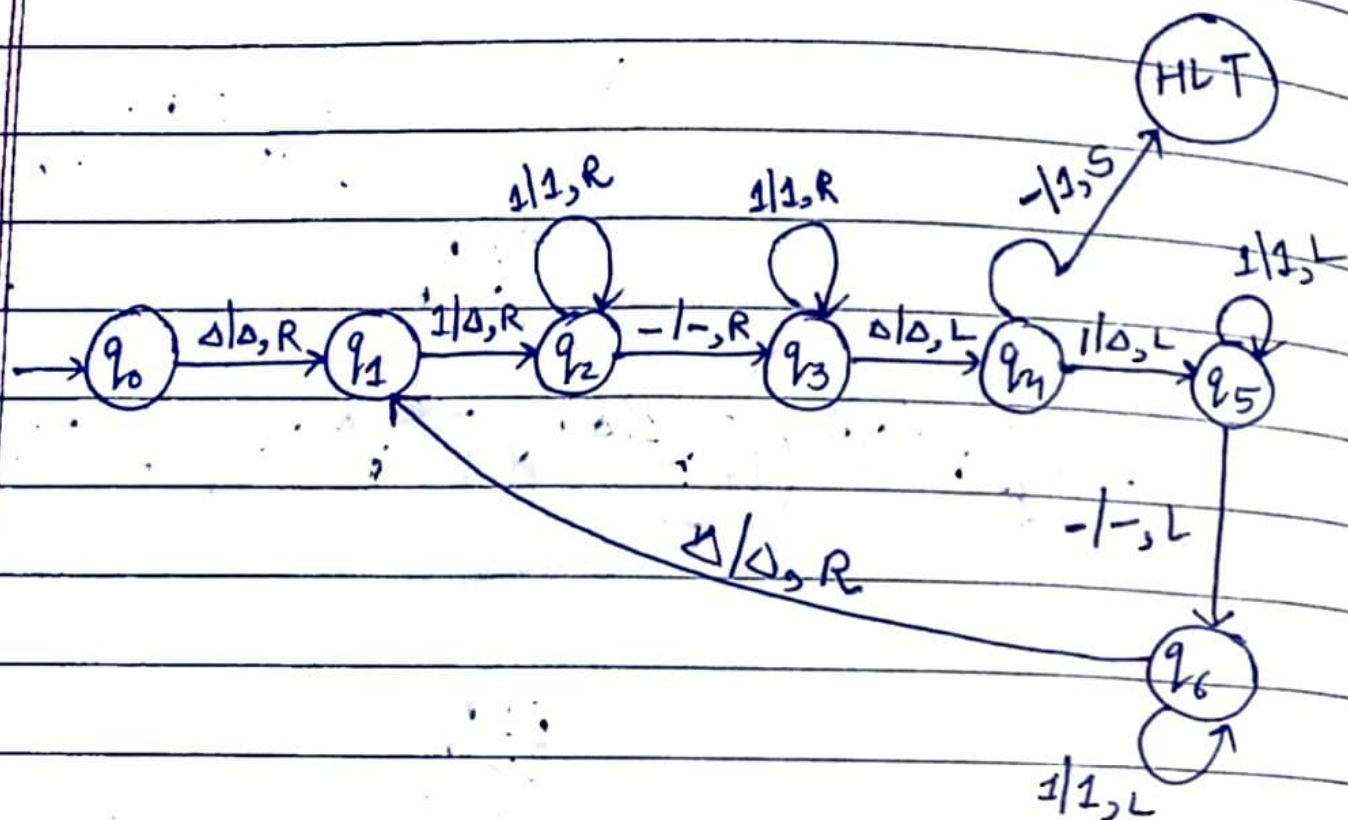
→ Replace + with Δ and remove one from the end.



(12) Design TM for unary subtraction:
 $\Delta 1111 \div 11 \Delta = \Delta 11 \Delta$

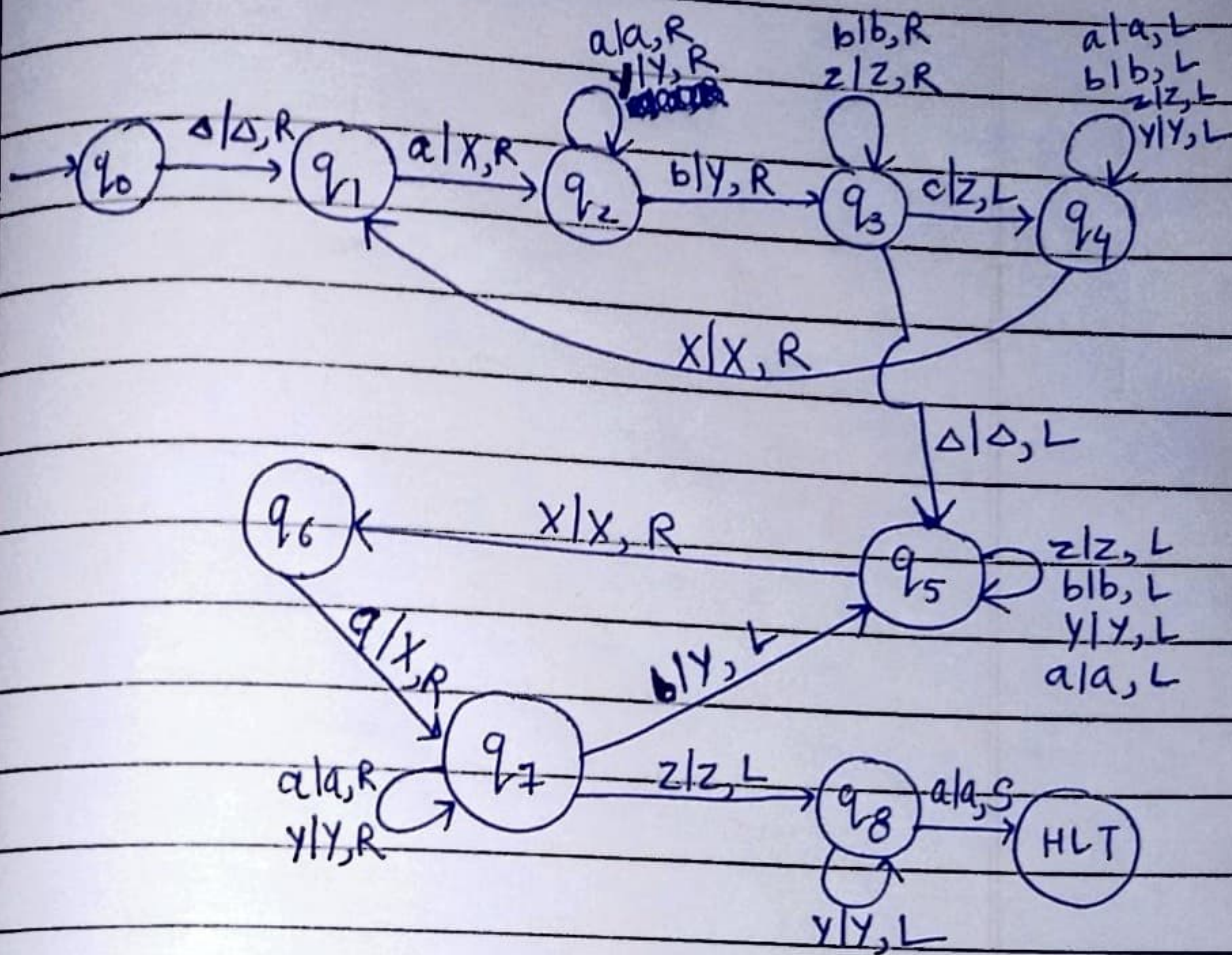
OR

Design TM for unary subtraction
 $f(a-b) = a > b$.

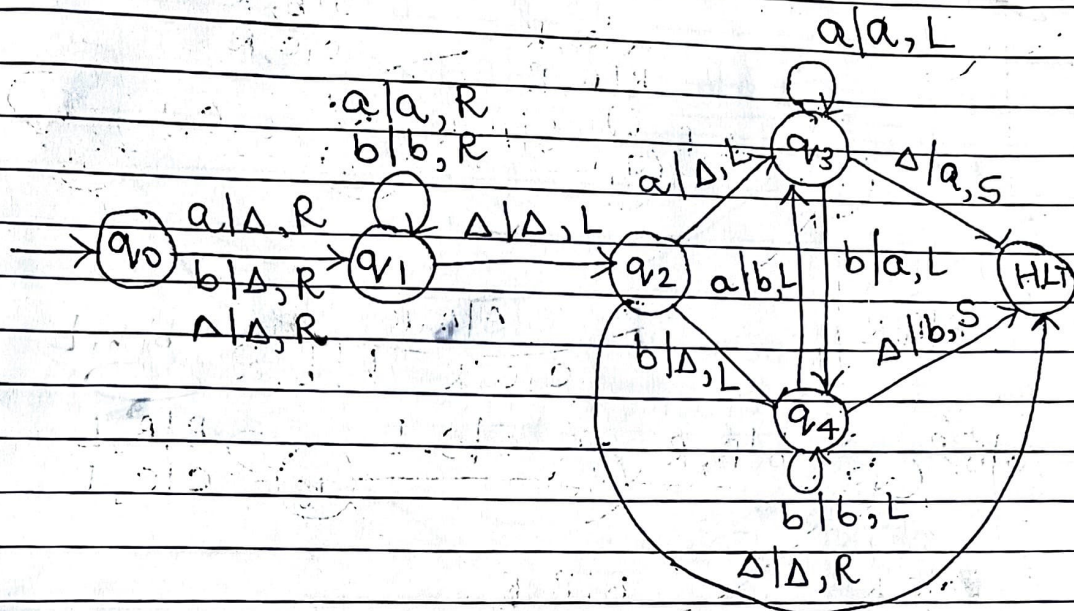


(13) Design TM for $L = \{a^i b^j c^k \mid i \geq j \geq k\}$
 $i, j, k \geq 1$.

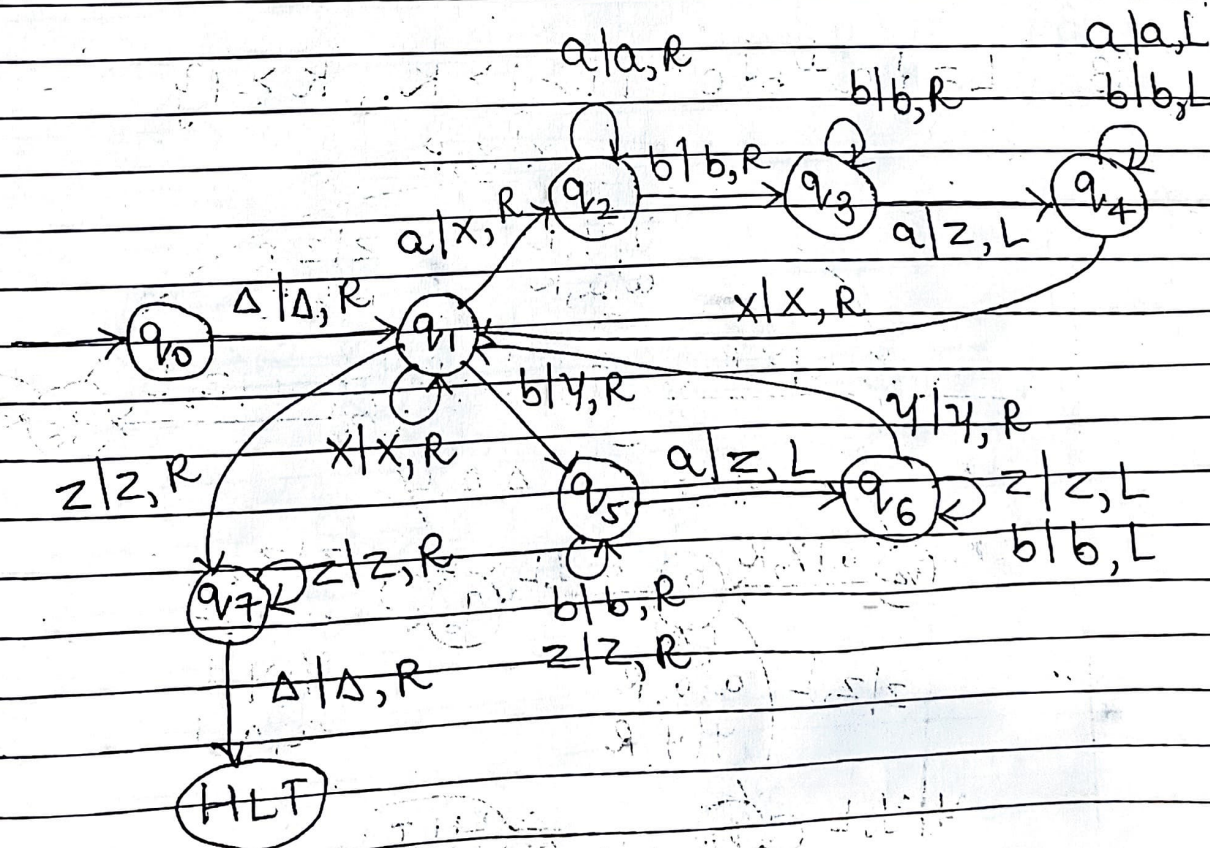
$L = \{ \Delta a a a b b b c c \Delta \}$



10. Construct a TM to delete a symbol from a string from $\Sigma = \{a, b\}$



11. $L = \{a^n b^m a^{n+m} \mid n, m \geq 1\}$



$\Delta \Delta \Delta \Delta \Delta$
 $\Delta \Delta \Delta \Delta \Delta$
 $\Delta \Delta \Delta$

$\rightarrow L = \{ a^i b^j c^k \mid i+j=k, i, j, k \geq 1 \}$

